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Abstract

An anchor-handling device comprises a housing shaped to define a channel having an anchor-storage zone and a proximal opening. A tissue anchor is stored in the anchor-storage zone. The anchor-handling device is configured such that, while the tissue anchor is stored in the anchor-storage zone, the tissue anchor is movable out of the anchor-storage zone toward the proximal opening in response to a proximally-directed force being applied to the tissue anchor. The anchor-handling device comprises an element that serves as an indicator of movement of the tissue anchor out of the anchor-storage zone toward the proximal opening by the element moving in response to the proximally-directed force. Other embodiments are also described.

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8419825	12/2012	Burgler et al.	N/A	N/A

8430926	12/2012	Kirson	N/A	N/A
8449573	12/2012	Chu	N/A	N/A
8449599	12/2012	Chau et al.	N/A	N/A
8454686	12/2012	Alkhatib	N/A	N/A
8460370	12/2012	Zakay	N/A	N/A
8460371	12/2012	Hlavka et al.	N/A	N/A
8475491	12/2012	Milo	N/A	N/A
8475525	12/2012	Maisano et al.	N/A	N/A
8480732	12/2012	Subramanian	N/A	N/A
8518107	12/2012	Tsukashima et al.	N/A	N/A
8523940	12/2012	Richardson et al.	N/A	N/A
8551161	12/2012	Dolan	N/A	N/A
8585755	12/2012	Chau et al.	N/A	N/A
8591576	12/2012	Hasenkam et al.	N/A	N/A
8608797	12/2012	Gross et al.	N/A	N/A
8628569	12/2013	Benichou et al.	N/A	N/A
8628571	12/2013	Hacohen et al.	N/A	N/A
8641727	12/2013	Starksen et al.	N/A	N/A
8652202	12/2013	Alon et al.	N/A	N/A
8652203	12/2013	Quadri et al.	N/A	N/A
8679174	12/2013	Ottma et al.	N/A	N/A
8685086	12/2013	Navia et al.	N/A	N/A
8728097	12/2013	Sugimoto et al.	N/A	N/A
8728155	12/2013	Montorfano et al.	N/A	N/A
8734467	12/2013	Miller et al.	N/A	N/A
8734699	12/2013	Heideman et al.	N/A	N/A
8740920	12/2013	Goldfarb et al.	N/A	N/A
8747463	12/2013	Fogarty et al.	N/A	N/A
8778021	12/2013	Cartledge	N/A	N/A
8784481	12/2013	Alkhatib et al.	N/A	N/A
8790367	12/2013	Nguyen et al.	N/A	N/A
8790394	12/2013	Miller et al.	N/A	N/A
8795298	12/2013	Hernlund et al.	N/A	N/A
8795355	12/2013	Alkhatib	N/A	N/A
8795356	12/2013	Quadri et al.	N/A	N/A
8795357	12/2013	Yohanan et al.	N/A	N/A
8808366	12/2013	Braido et al.	N/A	N/A
8808368	12/2013	Maisano et al.	N/A	N/A
8845717	12/2013	Khairkhahan et al.	N/A	N/A
8845723	12/2013	Spence et al.	N/A	N/A
8852261	12/2013	White	N/A	N/A
8852272	12/2013	Gross et al.	N/A	N/A
8858623	12/2013	Miller et al.	N/A	N/A
8864822	12/2013	Spence et al.	N/A	N/A
8870948	12/2013	Erzberger et al.	N/A	N/A
8870949	12/2013	Rowe	N/A	N/A
8888843	12/2013	Khairkhahan et al.	N/A	N/A
8889861	12/2013	Skead et al.	N/A	N/A
8894702	12/2013	Quadri et al.	N/A	N/A
8911461	12/2013	Traynor et al.	N/A	N/A

8911494	12/2013	Hammer et al.	N/A	N/A
8926696	12/2014	Cabiri et al.	N/A	N/A
8926697	12/2014	Gross et al.	N/A	N/A
8932343	12/2014	Alkhatib et al.	N/A	N/A
8932348	12/2014	Solem et al.	N/A	N/A
8940044	12/2014	Hammer et al.	N/A	N/A
8945211	12/2014	Sugimoto	N/A	N/A
8951285	12/2014	Sugimoto et al.	N/A	N/A
8951286	12/2014	Sugimoto et al.	N/A	N/A
8961595	12/2014	Alkhatib	N/A	N/A
8961602	12/2014	Kovach et al.	N/A	N/A
8979922	12/2014	Jayasinghe et al.	N/A	N/A
8992604	12/2014	Gross et al.	N/A	N/A
9005273	12/2014	Salahieh et al.	N/A	N/A
9011520	12/2014	Miller et al.	N/A	N/A
9011530	12/2014	Reich et al.	N/A	N/A
9023100	12/2014	Quadri et al.	N/A	N/A
9072603	12/2014	Tuval et al.	N/A	N/A
9107749	12/2014	Bobo et al.	N/A	N/A
9119719	12/2014	Zipory et al.	N/A	N/A
9125632	12/2014	Loulmet et al.	N/A	N/A
9125742	12/2014	Yoganathan et al.	N/A	N/A
9138316	12/2014	Bielefeld	N/A	N/A
9173646	12/2014	Fabro	N/A	N/A
9180005	12/2014	Lashinski et al.	N/A	N/A
9180007	12/2014	Reich et al.	N/A	N/A
9192472	12/2014	Gross et al.	N/A	N/A
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9226825	12/2015	Starksen et al.	N/A	N/A
9265608	12/2015	Miller et al.	N/A	N/A
9326857	12/2015	Cartledge et al.	N/A	N/A
9414921	12/2015	Miller et al.	N/A	N/A
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9474606	12/2015	Zipory et al.	N/A	N/A
9526613	12/2015	Gross et al.	N/A	N/A
9561104	12/2016	Miller et al.	N/A	N/A
9579090	12/2016	Simms et al.	N/A	N/A
9693865	12/2016	Gilmore et al.	N/A	N/A
9730793	12/2016	Reich et al.	N/A	N/A
9788941	12/2016	Hacohen	N/A	N/A
9801720	12/2016	Gilmore et al.	N/A	N/A
9907547	12/2017	Gilmore et al.	N/A	N/A
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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS (1) The present application is a Continuation of U.S. patent application Ser. No. 17/378,559 to Zipory et al., filed Jul. 16, 2021, and titled "Anchor Magazine"; which is a Continuation of U.S. patent application Ser. No. 16/239,279 to Zipory et al., filed Jan. 3, 2019, and titled "Anchor Magazine" (now U.S. Pat. No. 11,065,001); which is a Continuation of U.S. patent application Ser. No. 15/030,731 to Zipory et al., titled "Anchor Magazine" (now U.S. Pat. No. 10,299,793); which is the US National Phase of PCT application IL2014/050914 to Zipory et al., filed Apr. 20, 2016, which published as WO 2015/059699; which claims priority from US Provisional Patent Application U.S. 61/894,486 to Zipory et al., titled "Anchor Magazine", filed Oct. 23, 2013. Each of the above applications is incorporated herein by reference. (2) The present application is related to PCT Patent Application IL2013/050861 to Herman et al., titled "Percutaneous tissue anchor techniques", filed on Oct. 23, 2013, which published as WO 2014/064695, and PCT Patent Application IL2013/050860 to Sheps et al., titled "Controlled steering functionality for implant-delivery tool", filed on Oct. 23, 2013, which published as WO 2014/064694, both of which are incorporated herein by reference.

FIELD OF THE INVENTION

(1) The present invention relates in general to handling of tissue anchors. More specifically, the present invention relates to devices and techniques for handling of a plurality of tissue anchors, and the use thereof at a heart valve of a patient.

BACKGROUND

(2) Tissue anchors are placed intracorporeally so as to anchor implants to a tissue of a subject. Typically, this intracorporeal placement necessitates that the tissue anchors are small, e.g., having a greatest dimension (e.g., a length) of less than 11 mm and/or a maximum width of 3 mm. It is therefore typically advantageous to provide devices and techniques to facilitate handling of the tissue anchors.

SUMMARY OF THE INVENTION

(3) An anchor-handling device is configured to facilitate handling of one or more tissue anchors. The anchor-handling device retains the anchors within an anchor-storage zone of a channel defined by a housing until a tool such as an anchor driver is used to retrieve the anchor. The tool is advanced through the channel, coupled to the anchor, and removed proximally out of the channel with the anchor. The anchor-handling device is configured to release (e.g., dispense) the anchor only when a proximally-directed force applied by the tool to the anchor is greater than a pre-defined threshold force (i.e., is sufficient), so as to prevent inadvertent exit of the anchor.

(4) A retaining member is configured to retain the tissue anchor in the anchor-storage zone, typically by obstructing exit of the tissue anchor. The sufficient proximally-directed force moves the retaining member out of the way of the anchor, e.g., by moving the anchor to push the retaining member out of the way. Typically, an inhibitor inhibits movement of the retaining member, thereby configuring the retaining member to move out of the way of the anchor only in response to the sufficient proximally-directed force.

(5) For some applications, the anchor-handling device is used in combination with a multi-component tubular system for transcatheter delivery of an implant, e.g., to facilitate sequential delivery of a plurality of anchors to the implant via the system.

(6) There is therefore provided, in accordance with an application of the present invention, apparatus for use with an anchor driver, the apparatus including: a housing, shaped to define a channel having an anchor-storage zone and a proximal opening configured to provide access for the anchor driver to the anchor-storage zone; a tissue anchor, stored in the anchor-storage zone and slidable through the channel; and a retaining member: having a retaining state in which the retaining member is configured to retain the tissue anchor in the anchor-storage zone, and being configured, by moving in response to a proximally-directed force applied to the tissue anchor, to allow the tissue anchor to leave the anchor-storage zone in response to the proximally-directed

force, the proximally-directed force being greater than a pre-determined threshold force.

(7) In an application, the apparatus is configured such that after removal of the tissue anchor from the housing, a distally-directed force required to return the apparatus to the retaining state is more than twice as great as the threshold force.

(8) In an application, the retaining member is configured such that the threshold force is 300-1500 grams force.

(9) In an application, the tissue anchor has a mass, and the retaining member is configured such that the threshold force, measured in grams force, is 1000-150,000 times greater than the mass of the tissue anchor, measured in grams.

(10) In an application, the apparatus further includes an inhibitor, configured to configure the retaining member to (i) retain the tissue anchor in the anchor-storage zone, and (ii) to allow the tissue anchor to leave the anchor-storage zone in response to the proximally-directed force.

(11) In an application, the tissue anchor is dimensioned to fit snugly in the anchor-storage zone.

(12) In an application, the apparatus further includes a multi-component tubular system for transcatheter implantation of an implant into a subject, the implant configured to be anchored to tissue of the subject using the tissue anchor, and the housing being coupled to a component of the multi-component tubular system.

(13) In an application, the component of the multi-component tubular system includes a stand, and the housing is coupled to the stand.

(14) In an application, the multi-component tubular system defines a proximal port through which the anchor is introducible, and the housing is coupled to the component of the multi-component tubular system such that the proximal opening of the housing is disposed between 1 and 40 cm from the port of the multi-component tubular system.

(15) In an application, the multi-component tubular system defines a proximal port through which the anchor is introducible, and the housing is coupled to the component of the multi-component tubular system such that the proximal opening of the housing faces generally the same direction as the port of the multi-component tubular system.

(16) In an application: the housing is configured to define a plurality of channels, each of the plurality of channels having a respective anchor-storage zone and a respective proximal opening, and the apparatus includes a plurality of tissue anchors, slidable through a respective channel and configured to be stored in a respective anchor-storage zone.

(17) In an application, the apparatus includes a plurality of retaining members, each retaining member configured to retain a respective tissue anchor in the respective anchor-storage zone, and to allow the respective tissue anchor to leave the respective anchor-storage zone in response to a proximally-directed force applied to the respective tissue anchor.

(18) In an application, in the retaining state, at least a portion of the retaining member obstructs proximal movement of the tissue anchor by being disposed within the channel.

(19) In an application, the apparatus further includes the anchor driver, and in the retaining state, the anchor driver is slidable through the channel and lockable to the tissue anchor while at least the portion of the retaining member obstructs proximal movement of the tissue anchor by being disposed within the channel.

(20) In an application, in the retaining state, the anchor driver is slidable through the channel such that a part of the anchor driver becomes positioned between a part of the tissue anchor and a part of the retaining member, and the anchor driver is lockable to the tissue anchor only while the part of the anchor driver is positioned between the part of the tissue anchor and the part of the retaining member.

(21) In an application: the tissue anchor includes a core, a tissue-engaging member coupled to a distal side of the core, and a coupling head coupled to a proximal side of the core, and in the retaining state, at least the portion of the retaining member that obstructs the proximal movement of the tissue anchor obstructs the proximal movement of the tissue anchor by engaging the core.

- (22) In an application: the housing is shaped to define a chamber that is in fluid communication with the channel, at least part of the retaining member is configured to slide within the chamber in response to the proximally-directed force applied to the tissue anchor.
- (23) In an application, a first end of the chamber is in fluid communication with the channel, the housing defines a chamber opening of the chamber at a second end of the chamber, the portion of the retaining member includes a first portion of the retaining member, and the retaining member is configured such that, in response to the proximally-directed force applied to the tissue anchor, a second portion of the retaining member moves out of the chamber opening.
- (24) In an application, the retaining member is configured such that, in response to the proximally-directed force applied to the tissue anchor, a second portion of the retaining member moves out of the housing.
- (25) In an application, the apparatus is configured such that after removal of the tissue anchor from the housing, a distally-directed force required to return the apparatus to the retaining state is more than twice as great as the threshold force.
- (26) In an application, the retaining member includes a pin, configured to slide through the chamber.
- (27) In an application: the housing is shaped to define a cavity that is in fluid communication with the chamber, at least a portion of the retaining member is resilient, the retaining member is shaped to define a detent, in the retaining state, the resilience of at least the portion of the retaining member holds the detent within the cavity, and the retaining member is configured to deform in response to the proximally-directed force applied to the tissue anchor, such that the detent exits the cavity.
- (28) In an application: the cavity includes a first cavity, the housing is shaped to define a second cavity that is in fluid communication with the chamber, and the apparatus is dimensioned such that when the retaining member allows the tissue anchor to leave the anchor-storage zone, further proximal movement of the retaining member causes the detent to move into the second cavity.
- (29) In an application, the second cavity is larger in at least one dimension than the first cavity.
- (30) In an application, the second cavity is differently shaped to the first cavity.
- (31) In an application, the second cavity and the detent are dimensioned such that when the detent is disposed within the second cavity, a distally-directed force required to return the apparatus to the retaining state is more than twice as great as the threshold force.
- (32) In an application, at least a portion of the pin is dimensioned to slide snugly through the chamber.
- (33) In an application, the apparatus further includes an inhibitor tongue having a pin-contacting portion that is in contact with the pin, and configured to (i) inhibit the pin from sliding through the chamber in response to a sub-threshold force, and (ii) to allow the pin to slide through the chamber in response to the proximally-directed force applied to the tissue anchor.
- (34) In an application: the pin is shaped to define a cavity, at least a portion of the inhibitor tongue is resilient, in the retaining state, the resilience of at least the portion of the inhibitor tongue holds the pin-contacting portion within the cavity, and the inhibitor tongue is configured to deform in response to the proximally-directed force applied to the tissue anchor, such that the pin-contacting portion exits the cavity.
- (35) In an application: the cavity includes a first cavity. the pin is shaped to define a second cavity, the apparatus is dimensioned such that when the retaining member allows the tissue anchor to leave the anchor-storage zone, further proximal movement of the retaining member causes the pin-contacting portion to move into the second cavity.
- (36) In an application, the second cavity is larger in at least one dimension than the first cavity.
- (37) In an application, the second cavity is differently shaped to the first cavity.
- (38) In an application, the second cavity and the pin-contacting portion are dimensioned such that when the pin-contacting portion is disposed within the second cavity, a distally-directed force

required to return the apparatus to the retaining state is more than twice as great as the threshold force.

(39) In an application, the chamber is in fluid communication with the channel at a distal end of the chamber, and has a proximal-distal longitudinal axis that is disposed at between 5 and 30 degrees from a proximal-distal longitudinal axis of the channel.

(40) In an application, the proximal-distal longitudinal axis of the chamber is disposed at between 5 and 20 degrees from the proximal-distal longitudinal axis of the channel.

(41) In an application, the proximal-distal longitudinal axis of the chamber is disposed at between 11 and 14 degrees from the proximal-distal longitudinal axis of the channel.

(42) In an application, a central longitudinal axis of the chamber is parallel with a central longitudinal axis of the channel.

(43) In an application, the tissue anchor is dimensioned to fit snugly through the channel.

(44) In an application, the tissue anchor includes a core, a tissue-engaging member coupled to a distal side of the core, and a coupling head, the core is dimensioned to fit snugly through the channel, and the tissue-engaging member is dimensioned so as to not touch the housing when the tissue anchor moves through the channel.

(45) In an application, the apparatus further includes the anchor driver.

(46) In an application, the anchor driver includes: at a distal end thereof, an anchor-engaging head introducible through the opening of the housing and actuatable to be reversibly coupled to the tissue anchor; at a proximal end thereof, a handle including an adjuster configured to actuate the anchor-engaging head; and a flexible shaft: disposed between the distal end of the anchor driver and the proximal end of the anchor driver, having a length of 50-250 cm, and configured to be transcatheterally advanced through vasculature of a subject.

(47) In an application, the opening of the housing is rotationally asymmetrical, a transverse cross-section of the anchor-engaging head is rotationally asymmetrical, and the opening limits a range of rotational orientations of the anchor-engaging head with respect to the opening in which the anchor-engaging head is introducible through the opening.

(48) In an application, the opening of the housing and the transverse cross-section of the anchor-engaging head each have the shape of an ellipse that has had a segment removed.

(49) In an application, the tissue anchor is stored in the anchor-storage zone in a given rotational orientation of the tissue anchor with respect to the opening, the anchor-engaging head is couplable to the tissue anchor in not all rotational orientations of the head with respect to the tissue anchor, and the anchor-engaging head is couplable to the tissue anchor without rotating the anchor-engaging head subsequently to introducing the anchor-engaging head through the opening.

(50) In an application, the opening limits the range of rotational orientations such that the anchor-engaging head is introducible through the opening in only a given rotational orientation of the head with respect to the opening.

(51) In an application, the apparatus further includes a base, and: the housing is couplable to the base, the base is configured to at least partly immobilize the housing, and the base is shaped to define a receptacle for housing and at least partly immobilizing the handle.

(52) In an application, when the housing is coupled to the base, the housing is disposed less than 30 cm from the receptacle.

(53) In an application: the receptacle is a handle receptacle, the housing is reversibly couplable to the base, the base is shaped to further define a housing receptacle, configured to house the housing, the base further includes a locking element, movable between a locked state that locks the housing within the receptacle, and an unlocked state that facilitates release of the housing from the receptacle.

(54) In an application, the adjuster is operable while the receptacle houses the handle.

(55) In an application, the apparatus is configured such that while the receptacle houses the handle, a human operator may: with a first hand of the operator, grasp a distal portion of the driver and

introduce the head into the opening, and with a second hand of the operator, reversibly actuate the head by operating the adjuster while grasping the distal portion of the driver with the first hand.

(56) There is further provided, in accordance with an application of the present invention, apparatus for use with an anchor driver, the apparatus including: a housing, shaped to define a channel having (a) an anchor-storage zone and (b) a proximal opening configured to provide access for the anchor driver to the anchor-storage zone; a tissue anchor, slidable through the channel and configured to be stored in the anchor-storage zone; and a retaining member: having a retaining state in which the retaining member is configured to retain the tissue anchor in the anchor-storage zone, and being disposed within the housing such that sliding of the tissue anchor proximally out of the anchor-storage zone and through the channel causes the retaining member to slide in an at least partly proximal direction.

(57) In an application, the retaining member is disposed within the housing such that sliding of the tissue anchor proximally out of the anchor-storage zone and through the channel causes the retaining member to slide along an axis that is disposed at an angle of less than 30 degrees with respect to a central longitudinal axis of the channel.

(58) There is further provided, in accordance with an application of the present invention, apparatus, including: a housing, shaped to define a channel having an anchor-storage zone and a proximal opening; an anchor driver including an anchor-engaging head, a handle, and a shaft therebetween, and: the shaft is flexible and is configured to be transluminally advanced into a subject, and the anchor-engaging head is dimensioned to be advanceable through the proximal opening toward the anchor-storage zone; and a tissue anchor: stored in the anchor-storage zone, including (i) a coupling head configured to be locked to the anchor-engaging head while the tissue anchor is in the anchor-storage zone, and (ii) a tissue-engaging member configured to be driven into tissue of the subject using the anchor driver, and configured such that, while stored in the anchor-storage zone, the tissue anchor is movable out of the anchor-storage zone toward the proximal opening only in response to a proximally-directed force being applied to the tissue anchor, the proximally-directed force being greater than a pre-determined threshold force.

(59) In an application, the tissue anchor is configured to be movable out of the anchor-storage zone only in response to the proximally-directed force, by being dimensioned with respect to at least one dimension of the housing such that the tissue anchor is movable out of the anchor-storage zone only in response to the proximally-directed force.

(60) There is further provided, in accordance with an application of the present invention, apparatus, including: an anchor-handling device including a housing, shaped to define a channel having an anchor-storage zone and a proximal opening; an anchor driver including an anchor-engaging head, a handle, and a shaft therebetween, and: the shaft is flexible and is configured to be transluminally advanced into a subject, and the anchor-engaging head is dimensioned to be advanceable through the proximal opening toward the anchor-storage zone; and a tissue anchor: stored in the anchor-storage zone, including (i) a coupling head configured to be locked to the anchor-engaging head while the tissue anchor is in the anchor-storage zone, and (ii) a tissue-engaging member configured to be driven into tissue of the subject using the anchor driver, and configured such that, while stored in the anchor-storage zone, the tissue anchor is movable out of the anchor-storage zone toward the proximal opening in response to a proximally-directed force being applied to the tissue anchor by the anchor driver, and the anchor-handling device is configured to provide an indication of the movement of the tissue anchor out of the anchor-storage zone toward the proximal opening in response to the proximally-directed force.

(61) In an application, the anchor-handling device is configured to provide the indication by the housing being at least in part transparent, such that the movement of the tissue anchor is viewable from outside the housing.

(62) In an application, the anchor-handling device is configured to provide the indication by including an element that moves with respect to the housing in response to the movement of the

tissue anchor.

(63) In an application, the element that moves with respect to the housing moves out of the housing in response to the movement of the tissue anchor.

(64) In an application, the element that moves with respect to the housing moves with respect to the housing at a rate that is relative to a rate at which the anchor moves with respect to the housing.

(65) The present invention will be more fully understood from the following detailed description of applications thereof, taken together with the drawings, in which:

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. 1A-F are schematic illustrations of an anchor-handling device, configured to facilitate handling of at least one tissue anchor, in accordance with some applications of the invention;

(2) FIG. 2 is a schematic illustration of a multiple-anchor-handling device, in accordance with some applications of the invention;

(3) FIGS. 3A-C are schematic illustrations of a system for transcatheter delivery of an implant, and anchoring of the implant using an anchor driver and a plurality of anchors provided in the multiple-anchor-handling device of FIG. 2, in accordance with some applications of the invention;

(4) FIGS. 4A-F are schematic illustrations of an anchor-handling device, configured to facilitate handling of at least one tissue anchor, in accordance with some applications of the invention; and

(5) FIGS. 5A-C are schematic illustrations of a base to which an anchor-handling device is couplable, and which is configured to at least partly immobilize the anchor-handling device, in accordance with some applications of the invention.

DETAILED DESCRIPTION OF EMBODIMENTS

(6) Reference is made to FIGS. 1A-F, which are schematic illustrations of an anchor-handling device **20**, configured to facilitate handling of at least one tissue anchor **40**, in accordance with some applications of the invention. Device **20** comprises a housing **22** that defines a channel **24**, an anchor-storage zone **26** (e.g., at a distal end of the channel) and an opening **28** (e.g., at a proximal end of the channel) that provides access to the channel and the anchor-storage zone. Typically, there is a smooth transition between anchor-storage zone **26** and channel **24**. Device **20** further comprises a retaining member, such as a pin **30**, which is configured to retain tissue anchor **40** in zone **26**, and to stop retaining the tissue anchor in response to a sufficient proximally-directed force applied to the tissue anchor. That is, when a proximally-directed force that is greater than a pre-determined threshold force is applied to tissue anchor **40**, the retaining member stops retaining (e.g., releases) the tissue anchor.

(7) Typically, the retaining member (e.g., pin **30**) has a retaining state in which it retains tissue anchor **40** within zone **26**, and is moved out the retaining state when the sufficient proximally-directed force is applied to the tissue anchor. FIG. 1A shows, in accordance with some applications of the invention, pin **30** in a retaining state thereof, in which at least a portion **29** (e.g., an obstructing portion, and/or a distal portion) of the pin is disposed within channel **24** (e.g., proximal to anchor **40**), thereby retaining the anchor in zone **26** by obstructing proximal movement of the tissue anchor. Typically, portion **29** obstructs proximal movement of anchor **40** by engaging and/or obstructing core **41** of the anchor (described hereinbelow).

(8) It is to be noted that although pin **30** is shown as being generally cylindrical (i.e., having a generally circular transverse cross-section), the term “pin”, as used throughout the present application, including the specification and the claims, may include a pin having a different shape (e.g., having a noncircular transverse cross-section). For example, pin **230** (described hereinbelow with reference to FIGS. 4A-5C) typically has a rectangular cross-section.

(9) FIG. 1A shows an anchor driver **60** being advanced toward opening **28** of channel **24**, and FIG.

1B shows the anchor driver having been further advanced into and through channel 24, to anchor 40. As more clearly shown in FIGS. 1A and 1F, opening 28 is typically beveled (i.e., disposed at an angle smaller than 90 degrees to a longitudinal axis of channel 24), such that the opening has a greater area than does a transverse cross section of the channel, thereby facilitating introduction of driver 60 into the channel. It is to be noted that the shape of opening 28 provides a proximal region 25 in which channel 24 is at least half open on a lateral side (i.e., at least half of the circumferential surface of the channel is missing). This shape thereby facilitates placement of a driver head 62 of driver 60 in proximal region 25 of channel 24, e.g., by reducing a requirement for the driver to be aligned with the channel before its introduction into the channel. Driver 60 (e.g., head 62 thereof) can subsequently be advanced further into channel 24, using region 25 as a guide track.

(10) Driver 60 typically comprises an anchor-engaging head 62 at a distal end of the driver, and a shaft 64 proximal to the anchor-engaging head. Shaft 64 is flexible and advanceable (e.g., transcatheterally) through vasculature of a subject, and typically has a length greater than 20 cm, and/or less than 2.5 m, such as greater than 50 cm and/or less than 1.5 m, e.g., between 0.9 m and 1.2 m. For some applications, driver 60 comprises a handle 66 at a proximal end of shaft 64, the handle comprising an adjuster 68 (e.g., a switch or a lever) configured to actuate engaging head 62.

(11) Tissue anchor 40 typically comprises a core 41, a tissue-engaging member 44 coupled to a distal side of the core, and a coupling head 42 coupled to a proximal side of the core. Engaging head 62 is configured to be reversibly couplable to tissue anchor 40 (e.g., to coupling head 42 thereof), so as to facilitate acquisition of the anchor from device 20, driving of the anchor into tissue of the subject, and subsequent release of the anchor and withdrawal of driver 60 from the subject. For example, actuation of engaging head 62 by adjuster 68 may comprise transitioning the engaging head between (i) an open state in which the engaging head is configured to receive and/or release anchor 40 (FIGS. 1A-B), and (ii) a closed state in which the engaging head, having received the anchor (FIG. 1C), is coupled (e.g., locked) to the anchor.

(12) FIG. 1B shows engaging head 62 having received anchor 40, but not yet coupled (e.g., locked) to the anchor. It is to be noted that in this position part of driver 60 (e.g., head 62) is disposed (e.g., sandwiched) between part of the retaining member (e.g., portion 29) and part of the anchor (e.g., coupling head 42).

(13) FIG. 1C shows engaging head 62 being coupled (e.g., locked) to anchor 40. For some applications of the invention, engaging head 62 comprises a detent 70 that is transitioned into the closed state when a controller, such as a rod or a wire 72, is moved distally by adjuster 68, and automatically transitions back into the open state when the wire is withdrawn. FIG. 1C shows wire 72 having been moved distally into engaging head 62, and detent 70 having been pushed into the closed state, thereby coupling the engaging head to anchor 40 (e.g., to coupling head 42 thereof).

(14) FIGS. 1D-E show anchor 40 being withdrawn proximally from zone 26 of channel 24 by the sufficient proximally-directed force being applied to the anchor by driver 60. As described hereinabove, in response to the sufficient proximally-directed force applied to the tissue anchor (i.e., if the proximally-directed force is greater than the pre-determined threshold force), the retaining member (e.g., pin 30) stops retaining the tissue anchor in zone 26. For example, and as shown in FIGS. 1D-E, the sufficient proximally-directed force overcomes the retention provided by pin 30 and pushes portion 29 of pin 30 out of the channel and into a chamber 74 that is in fluid communication with the channel (at least part of pin 30 thereby sliding within the chamber).

(15) It is hypothesized that this configuring of device 20 to require that the sufficient proximally-directed force be applied to tissue anchor 40 prevents inadvertent movement and/or exit of the tissue anchor (e.g., due to general transport or handling of the device), and/or withdrawal of the anchor by driver 60 when the driver is sub-optimally coupled to the anchor.

(16) For some applications, a first end 76 of chamber 74 is in fluid communication with channel 24, housing 22 defines an opening 78 at a second end of the chamber, and the pushing of portion 29 of pin 30 by the sufficient proximally-directed force pushes a second (e.g., proximal) portion 31 of

the pin out of opening 78. This feature and advantages thereof are described in more detail hereinbelow. Typically, chamber 74 has a proximal-distal longitudinal axis that is disposed at between 5 and 30 degrees, e.g., 5-20 degrees (e.g., 5-15 degrees or 10-degrees, such as between 11 and 14 degrees) with respect to the longitudinal axis of channel 24. It is hypothesized that, for some applications, this angular disposition of the channel and chamber facilitates the above described movement of pin 30 in response to the sufficient proximally-directed force applied to the tissue anchor.

(17) FIG. 1F shows anchor 40 having been fully withdrawn out of channel 24 via opening 28. Once anchor 40 has been fully withdrawn, driver 60 may be used to anchor tissue anchor 40 to tissue of a subject, e.g., by driving tissue-engaging member 44 (FIG. 1A) of the anchor into the tissue. It may be used as a tissue anchor as is known in the art. For example, using driver 60, anchor 40 may be advanced through a transluminal implant-delivery system and used to couple an implant to tissue of a subject, e.g., as described hereinbelow with reference to FIGS. 3A-C.

(18) For some applications, device 20 comprises an inhibitor, configured to configure the retaining member (e.g., pin 30) to (i) retain the tissue anchor in anchor-storage zone 26, and (ii) to stop retaining the tissue anchor in response to the sufficient proximally-directed force. For example, the inhibitor may comprise an inhibitor tongue 80, that has a pin-contacting portion 82 (e.g., a pin-contacting surface) that is in contact with pin 30, and that provides resistance that (i) inhibits sliding of the pin through chamber 74 (e.g., prevents sliding of the pin in response to an insufficient proximally-directed force, i.e., a proximally-directed force that is less than the pre-determined threshold force), and (ii) allows sliding of the pin through the chamber in response to the sufficient proximally-directed force that is greater than the pre-determined threshold force being applied to tissue anchor 40. Pin-contacting portion 82 is typically held in contact with pin 30 by a spring mechanism. For example, and as shown in FIGS. 1A-F, inhibitor tongue 80 may comprise an elastically-deformable (e.g., shape-memory) material, and may be coupled to housing 22 by in a manner in which the inhibitor tongue itself provides the spring mechanism.

(19) For some applications, pin 30 defines a cavity 32 therein (e.g., a recess or a notch in a lateral side of the pin), in which pin-contacting portion 82 is typically disposed while anchor is disposed within anchor-storage zone 26 (e.g., in a state in which the device is provided). For such applications, portion 82 serves as a detent. For such applications, cavity 32 and inhibitor tongue 80 are configured such that when a proximally-directed force equal to or greater than the threshold force is applied to anchor 40, pin 30 is pushed against pin-contacting portion 82, and inhibitor tongue 80 responsively deforms such that the pin-contacting portion moves out of cavity 32, allowing pin 30 to move further proximally (FIG. 1D). Typically, for such applications, once portion 82 has moved out of cavity 32, a proximally-directed force that is smaller than the threshold force is sufficient to move pin 30 further proximally. That is, once the initial resistance provided by the inhibitor is overcome, anchor 40 is further withdrawable using a smaller force than that required to overcome the initial resistance.

(20) It will be understood by those skilled in the art that it is possible to use other configurations to achieve a behavior similar to that described above. For example, housing 22 may define a cavity, and pin 30 may comprise a flexible protrusion that extends into the cavity of the housing.

(21) For some applications, the inhibitor (e.g., tongue 80) provides the resistance by applying friction against the retaining member (e.g., pin 30). For example, pin-contacting portion 82 may comprise a high-friction pin-contacting surface.

(22) Reference is made to FIG. 2, which is a schematic illustration of a multiple-anchor-handling device 100, in accordance with some applications of the invention. Device 100 defines a plurality of channels 24, each channel having a respective proximal opening 28 and a respective anchor-storage zone 26 that is configured to store a respective tissue anchor 40 (zone 26 and anchor 40 not visible in FIG. 2). Typically, device 100 further comprises a plurality of retaining members (e.g., pins 30), each retaining member being configured to retain a respective tissue anchor in its

respective anchor-storage zone **26**, and to stop retaining the respective tissue anchor in response to the sufficient proximally-directed force being applied to its respective tissue anchor. For some applications, device **100** comprises a plurality of devices **20**. For example, device **100** may comprise a plurality of housings **22**, each housing defining exactly one channel **24** and exactly one retaining member (e.g., pin **30**).

(23) As described hereinabove, for some applications, the sufficient proximally-directed force pushes a second (e.g., proximal) portion **31** of pin **30** out of opening **78** of chamber **74**. Therefore, when driver **60** is withdrawn proximally, movement of portion **31** toward and/or out of opening **78** indicates that anchor-engaging head **62** has been successfully coupled to tissue anchor **40**, and that the tissue anchor is also being withdrawn proximally. Thus, during an initial partial withdrawal of driver **60**, movement of portion **31** toward and/or out of opening **78** provides an indication to the operator (e.g., physician) to continue to withdraw driver **60**, whereas absence of such movement of portion **31** provides an indication to the operator to reattempt coupling of the driver to tissue anchor **40**.

(24) Following removal of anchor **40** from channel **24**, portion **31** remains exposed from opening **78**. This may be particularly useful for a physician using a multiple-anchor-handling device, such as device **100**, e.g., to prevent the physician inadvertently attempting to obtain an anchor from an empty zone **26**. That is, portion **31** functions as an empty-housing indicator.

(25) For some applications, pin **30** defines a second cavity **34** therein (e.g., a second notch in a lateral side of the pin), disposed closer to distal portion **29** than is cavity **32**. Second cavity **34** is positioned such that when (1) distal portion **29** is no longer obstructing anchor **40**, and (2) second portion **31** is exposed out of opening **78**, pin-engaging portion **82** of inhibitor tongue **80** moves into the second cavity (e.g., as shown in FIG. **1E**). In this state, tongue **80** inhibits pin **30** from moving distally back into housing **22**, thereby increasing the reliability of portion **31** functioning as an empty-housing indicator. For some applications, and as shown in FIGS. **1A-F**, second cavity **34** is shaped such that once portion **82** has moved into (e.g., engaged) second cavity **34**, the moving of portion **31** back into housing **22** requires a distally-directed force that is more than twice as great (in the opposite direction) as the threshold force that was previously required to move portion **31** out of the housing. For example, the moving of portion **31** back into housing **22** may be in effect prevented.

(26) Reference is made to FIGS. **3A-C**, which are schematic illustrations of a multi-component tubular system **120** for transcatheter delivery of an implant **140**, and anchoring of the implant using anchor driver **60** and a plurality of anchors **40** provided in multiple-anchor-handling device **100**, in accordance with some applications of the invention. Implant **140** comprises an annuloplasty structure comprising a sleeve **142**, a flexible elongated contracting member **144** that extends along the sleeve, and an adjustment mechanism **146** which facilitates contracting and expanding of the annuloplasty structure. Typically, adjustment mechanism **146** comprises a spool around which successive portions of member **144** are wound in order to contract the annuloplasty structure after implantation.

(27) Implant **140** is configured to be anchored to an annulus **10** of a valve of the heart **12** of a subject, such as a mitral valve **14** of the subject, and to change a dimension of the annulus when contracted or expanded using adjustment mechanism **146**.

(28) System **120** comprises one or more steerable catheters, and typically comprises an outer catheter **122** and an inner catheter **124** that is advanceable through the outer catheter. Outer catheter **122** is advanceable and steerable using a first handle **126**, and inner catheter **124** is advanceable and steerable using a second handle **128**. Typically, second handle **128** is couplable (e.g., lockable) to first handle **126**, e.g., after advancement of catheter **124** through catheter **122**.

(29) Implant **140** is typically (1) advanceable through inner catheter **124** in a delivery configuration in which adjustment mechanism **146** is disposed on an axis defined by sleeve **142**, distally to a distal end **143** of the sleeve, and (2) transitionable into an anchoring configuration in which the

adjustment mechanism is disposed laterally to the sleeve (FIG. 3A shows implant **140** in the anchoring configuration thereof). Advancement of implant **140** distally out of catheter **124** is typically controllable using a third handle **132** which is slidably coupled to handle **128**, e.g., via a handle-sliding track **138**.

(30) A portion of sleeve **142** (e.g., distal end **143**) is placed against annulus **10** (FIG. 3A). Typically, system **120** further comprises an implant-decoupling channel **130**, disposed within sleeve **142**, and slidable progressively proximally out of the sleeve, e.g., using a knob **134** coupled to a proximal end of channel **130**. Typically, channel **130** is used to hold the portion of sleeve **142** (e.g., distal end **143**) against annulus **10**.

(31) FIG. 3A shows anchor driver **60** (e.g., anchor-engaging head **62** thereof) being coupled to a first anchor **40** (not visible in FIG. 3A) which is disposed within a first housing **22** of device **100**, as described hereinabove (e.g., with reference to FIGS. 1A-C). Subsequently, as shown in FIG. 3B, anchor driver **60** is withdrawn from the first housing **22** of device **100**, while coupled to anchor **40**. Anchor **40** is then advanced into system **120** by advancing driver **60** (e.g., anchor-engaging head **62** and shaft **64** thereof) through the system (FIG. 3C). Typically, and as shown in FIG. 3C, the anchor is introduced into system **120** via a port **136** at a proximal end of handle **132**, and is slid through channel **130**, into sleeve **142**, and is screwed through sleeve **142** (e.g., distal end **143** thereof) and into annulus **10**.

(32) Driver **60** is subsequently removed from system **120**, coupled to a second anchor **40** disposed in a second housing **22** of device **100**, and reintroduced into the system. Channel **130** is withdrawn slightly proximally from the sleeve, and a second portion of the sleeve is held against a second site on annulus **10** before the second anchor is driven through sleeve **142**, anchoring the second portion of the sleeve to the second site. This process is repeated so as to place and anchor sleeve **142** around at least a portion of annulus **10**.

(33) Typically, and as shown in FIGS. 3A-C, device **100** is coupled to system **120** (e.g., the rest of system **120**) so as to facilitate access by driver **60** to openings **28** of channels **24** of housings **22**. For example, device **100** may be coupled to a stand (e.g., a base-plate) **121** of system **120**, and/or may be oriented such that openings **28** face generally the same direction as port **136** of system **120**, such that the operator (e.g., a physician) may easily move driver **60** between openings **28** and port **136**. Typically, device **100** is positioned such that openings **28** are closer than 1 m and/or greater than 1 cm (e.g., between 1 cm and 1 m, such as between 1 cm and 70 cm, such as between 1 cm and 40 cm) away from port **136** (or another opening through which anchors **40** are introduced into system **120**).

(34) Alternatively, device **100** may comprise a standalone unit, not coupled to system **120** or any other system.

(35) Subsequently, implant **140** may be adjusted (e.g., contracted) using an adjustment tool (not shown), advanceable over a guide member **141** to adjustment mechanism **146**.

(36) Reference is made to FIGS. 4A-F, which are schematic illustrations of an anchor-handling device **220**, configured to facilitate handling of at least one tissue anchor **40**, in accordance with some applications of the invention. Device **220** comprises a housing **222** that defines a channel **224** having an anchor-storage zone **226** and an opening **228** that provides access to the channel and the anchor-storage zone. Device **220** further comprises a retaining member, such as a pin **230**, which is configured to retain tissue anchor **40** in zone **226**, and to stop retaining the tissue anchor in response to a sufficient proximally-directed force applied to the tissue anchor. That is, when a proximally-directed force that is greater than a pre-determined threshold force is applied to tissue anchor **40**, the retaining member stops retaining (e.g., releases) the tissue anchor. For some applications FIGS. 4A-F, which show steps in the use of device **220**, generally correspond to FIGS. 1A-F, respectively, which show steps in the use of device **20**, mutatis mutandis.

(37) Typically, the retaining member (e.g., pin **230**) has a retaining state in which it retains tissue anchor **40** within zone **226**, and is moved out the retaining state when the sufficient proximally-

directed force is applied to the tissue anchor. FIG. 4A shows, in accordance with some applications of the invention, pin **230** in a retaining state thereof, in which at least a portion **229** (e.g., an obstructing portion) of the pin is disposed within channel **224** (e.g., proximal to anchor **40**), thereby retaining the anchor in zone **226** by obstructing proximal movement of the tissue anchor. Typically, portion **229** obstructs proximal movement of anchor **40** by engaging and/or obstructing core **41** of the anchor. Another portion of pin **230** is disposed in a chamber **274**, which is defined by housing **222** and is typically in fluid communication with channel **224** (e.g., in the absence of pin **230**). Typically, chamber **274** has a central longitudinal axis ax1 that is parallel with a central longitudinal axis ax2 of channel **24**.

(38) Similarly to device **20**, it is hypothesized that this configuring of device **220** prevents inadvertent movement and/or exit of the tissue anchor (e.g., due to general transport or handling of the device), and/or withdrawal of the anchor by driver **60** when the driver is sub-optimally coupled to the anchor.

(39) FIG. 4A shows anchor driver **60** (described hereinabove) being advanced toward opening **228** of channel **224**, and FIG. 4B shows the anchor driver having been further advanced into and through channel **224**, to anchor **40**, such that engaging head **62** has received anchor **40**, but not yet coupled (e.g., locked) to the anchor. It is to be noted that in this position part of driver **60** (e.g., head **62**) is disposed (e.g., sandwiched) between part of the retaining member (e.g., portion **229**) and part of the anchor (e.g., coupling head **42**).

(40) FIG. 4A shows respective cross-sections of opening **228** and driver head **62**, which are typically each rotationally asymmetrical. For some applications, and as shown, the cross-sections each have the shape of an ellipse (e.g., a circle) with a segment (e.g., a circular segment) removed. (For some applications opening **228** (e.g., the shape thereof) is defined partly by housing **222**, and partly by pin **230**.) Due to this rotational asymmetry, anchor-engaging head **62** is introducible through opening **228** in fewer than all rotational orientations of the head with respect to the opening. For example, head **62** may be introducible through opening **228** only in one or more particular rotational orientations (e.g., one particular orientation of the head) of the head with respect to the opening. This limitation of the rotational orientations in which head **62** may be introduced through opening **228** causes the head to be correctly rotationally oriented for coupling to coupling head **42** of anchor **40**, the anchor being stored in zone **226** in a given rotational orientation with respect to the opening. Therefore anchor-engaging head **62** is couplable to anchor **40**, without rotating the anchor-engaging head subsequently to introducing the anchor-engaging head through opening **228**.

(41) It is to be noted that this orientation-limitation may be applied to device **20**, mutatis mutandis, and that the lateral channel opening that provides region **25** of device **20** may be applied to device **220**, mutatis mutandis.

(42) FIG. 4C shows wire **72** having been moved distally into engaging head **62**, and detent **70** having been pushed into the closed state, thereby coupling the engaging head to anchor **40** (e.g., to coupling head **42** thereof).

(43) FIGS. 4D-E show anchor **40** being withdrawn proximally from zone **226** by the sufficient proximally-directed force being applied to the anchor by driver **60**. As described hereinabove, in response to the sufficient proximally-directed force applied to the tissue anchor (i.e., if the proximally-directed force is greater than the pre-determined threshold force), the retaining member (e.g., pin **230**) stops retaining the tissue anchor in zone **226**. For example, and as shown in FIGS. 4D-E, the sufficient proximally-directed force overcomes the retention provided by pin **230** such that at least a portion of pin **230** slides within chamber **274**, and at least a portion **231** (e.g., a proximal portion) of pin **230** moves out of housing **222**.

(44) FIG. 4D shows portion **231** of pin **230** beginning to move out of housing **222**, and FIG. 4E shows both portion **231** and anchor **40** disposed outside of the housing, such that pin **230** (e.g., portion **229** thereof) no longer obstructs anchor **40**. It is to be noted that for device **220**, the portion

229 of pin 230 that obstructs anchor 40 is disposed close to (e.g., within) the portion 231 of pin 230 that becomes exposed from housing 222, whereas for device 20 described hereinabove, the portion 29 of pin 30 that obstructs anchor 40 is disposed at another part (e.g., at the other end) of pin 30 from the portion 31 of pin 30 that becomes exposed from housing. Similarly, portion 229 typically becomes exposed from housing 222 upon withdrawal of anchor from device 20, whereas portion 29 typically remains within housing 22 upon withdrawal of anchor 40 from device 220.

(45) FIG. 4F shows anchor 40 having been fully removed from the housing. For some applications, to facilitate full disengagement of anchor 40 from pin 230, the anchor is moved slightly laterally with respect to pin 30. Once anchor 40 has been fully withdrawn, driver 60 may be used to anchor tissue anchor 40 to tissue of a subject, e.g., as described hereinabove.

(46) Device 220 comprises an inhibitor, configured to configure the retaining member (e.g., pin 30) to (i) retain the tissue anchor in anchor-storage zone 26, and (ii) to stop retaining the tissue anchor in response to the sufficient proximally-directed force. For example, pin 230 may comprise or define a detent 282 that, while anchor 40 is disposed within anchor-storage zone 226, is held by a spring mechanism within a cavity 232 (e.g., a notch) defined in chamber 274, and thereby serves as the inhibitor. For some applications at least a portion 280 of pin 230 is resilient, and thereby provides the spring mechanism. It is to be noted, however, that the scope of the invention includes the use of other spring mechanisms. The inhibitor provides resistance that (i) inhibits sliding of pin 230 through chamber 274, e.g., prevents sliding of the pin in the absence of a sufficient proximally-directed force (e.g., as shown in FIG. 4C), and (ii) stops inhibiting the sliding in response to the sufficient proximally-directed force by detent 282 moving out of cavity 232 (e.g., as shown in FIG. 4D). For example, and as shown, resilient portion 280 deforms (e.g., bends) in response to the sufficient proximally-directed force. This is typically facilitated by detent 282 and a proximal wall of cavity 232 having respective faces that are appropriately angled with respect to each other such that the proximally-directed force is converted into lateral movement of the detent out of the cavity. For example, and as shown, detent 282 may have a beveled edge.

(47) Typically, for such applications, once detent 282 has moved out of cavity 232, a proximally-directed force that is smaller than the threshold force is sufficient to move pin 30 further proximally. That is, once the initial resistance provided by the inhibitor is overcome, anchor 40 is further withdrawable using a smaller force than that required to overcome the initial resistance.

(48) (It will be understood by those skilled in the art that it is possible to use other configurations to achieve a behavior similar to that described above. For example, housing 222 may define a protrusion (e.g., a detent), and pin 30 may comprise a cavity (e.g., a notch) into which the protrusion extends.)

(49) For some applications, the inhibitor provides the resistance by applying friction against the wall of cavity 232. For example, pin 230 may have a high-friction wall-contacting surface.

(50) Device 220 is described hereinabove with reference to only one channel 224, zone 226, and restraining member. Typically however, the device defines a plurality of channels 224, each channel having a respective proximal opening 228 and a respective anchor-storage zone 226 that is configured to store a respective tissue anchor 40 (e.g., as described with reference to device 100, mutatis mutandis). Typically, device 220 further comprises a plurality of retaining members (e.g., pins 230), each retaining member being configured to retain a respective tissue anchor in its respective anchor-storage zone 226, and to stop retaining the respective tissue anchor in response to the sufficient proximally-directed force being applied to its respective tissue anchor.

(51) As described hereinabove, for some applications, the sufficient proximally-directed force pushes portion 231 of pin 30 out of housing 222. Therefore, as driver 60 is withdrawn proximally, movement of portion 231 out of housing 222 indicates that anchor-engaging head 62 has been successfully coupled to tissue anchor 40, and that the tissue anchor is also being withdrawn proximally. Thus, during an initial partial withdrawal of driver 60, movement of portion 231 out of housing 222 provides an indication to the operator (e.g., physician) to continue to withdraw driver

60, whereas absence of such movement of portion **231** provides an indication to the operator to reattempt coupling of the driver to tissue anchor **40**.

(52) Following removal of anchor **40** from channel **224**, portion **231** remains exposed outside of housing **222**. This may be particularly useful for a physician using device **220**, e.g., to prevent the physician inadvertently attempting to obtain an anchor from an empty anchor-storage zone **226**. That is, portion **231** functions as an empty-housing indicator.

(53) For some applications, a second cavity **234** is defined in chamber **274** (e.g., a second notch in a wall of the chamber), disposed more proximally with respect to housing **222** than is cavity **232**. Second cavity **234** is positioned such that when (1) portion **229** is no longer obstructing anchor **40**, and (2) portion **231** is exposed out of opening **278**, detent **282** (i.e., the inhibitor) moves into the second cavity (e.g., as shown in FIG. 4E). In this state, the detent inhibits pin **230** from moving distally back into housing **222**, thereby increasing the reliability of portion **31** functioning as an empty-housing indicator.

(54) For some applications, and as shown in FIGS. 1A-F, second cavity **234** is dimensioned and/or shaped such that once detent **282** has moved into (e.g., engaged) second cavity **234**, a distally-directed force required to return portion **231** into housing **222** (e.g., to return device **220** into its retaining state) is more than twice as great (in the opposite direction) as the threshold force that was previously required to move portion **231** out of the housing, and to remove anchor **40** from the housing. For example, and as shown, cavity **234** may be greater in one or more dimensions (e.g., wider and/or deeper) than cavity **232**. While detent **282** is disposed in cavity **232**, the beveled edge of the detent is partly exposed from cavity **232**, whereas while the detent is disposed in cavity **234**, the beveled edge is disposed entirely within the cavity. Other geometric configurations may also be used to generate this effect. For example, only a proximal face of detent **282** may be beveled.

(55) For some applications, housing **222** is at least in part transparent, so as to enable viewing of the coupling of driver **60** to anchor **40**, and/or withdrawal of the anchor from the housing.

(56) Reference is made to FIGS. 5A-C, which are schematic illustrations of a base **300** to which device **220** (e.g., housing **222** thereof) is couplable, and which is configured to at least partly immobilize the housing, in accordance with some applications of the invention. For some applications, and as shown, more than one device **220** is couplable to base **300**. Similarly to devices **20** and **100**, for some applications device **220** is used with multi-component tubular system **120** (described hereinabove). For such applications, anchor driver **60** (e.g., anchor-engaging head **62** thereof) is coupled to an anchor **40**, and then advanced into system **120** (e.g., via a port **136** at a proximal end of handle **132**). As described hereinabove, shaft **64** is flexible, and typically has a length greater than 20 cm, and/or less than 2.5 m, such as greater than 50 cm and/or less than 1.5 m, e.g., between 0.9 m and 1.2 m. Because of this flexibility and length, it may be difficult for an operator (e.g., a physician) to wield and operate handle driver **60** (e.g., to couple head **62** to an anchor while retaining control of shaft **64** and handle **66**). Base **300** facilitates such handling by defining a handle receptacle **302** for housing and at least partly immobilizing handle **66**.

(57) Typically, base **300** is configured such that when device **220** (e.g., housing **222** thereof) is coupled to the base, the housing is disposed less than 30 cm from receptacle **302** (e.g., less than 20 cm, e.g., less than 10 cm, such as less than 5 cm from the receptacle), and therefore less than 30 cm from handle **66** when the handle is disposed in the receptacle. Because shaft **64** is typically flexible, despite its length typically being greater than (e.g., more than twice as great, e.g., more than 5 times as great, such as 2-10 times as great as) the distance between device **220** and receptacle **302**, driver head **62** is insertable into device **220** while handle **66** is disposed in receptacle **302**.

(58) For some applications, device **220** is permanently coupled to base **300** (e.g., device **220** and base **300** may be integrated). For some applications device **220** is reversibly couplable to base **300**. For example, base **300** may further define at least one housing receptacle **304**, configured to house device **220**. For such applications, base **300** further comprises a locking element **306**, movable between a locked state that locks the housing within the receptacle, and an unlocked state that

facilitates release of the housing from the receptacle. For example, and as shown, locking element **306** may comprise one or more bars **308** that are advanceable through a portion of base **300** so as to protrude into (e.g., through) receptacle **304**; and housing **222** is shaped to define a respective one or more recesses **310** dimensioned to mate with the bars. Advancing bars **308** into receptacle **304** while housing **222** is disposed in the receptacle thereby locks the housing within the receptacle.

(59) Typically, receptacle **302** is dimensioned such that adjuster **68** is operable while the receptacle houses handle **66**. This, along with the proximity of handle **66** to device **220**, advantageously facilitates the operator (i) with a first hand, grasping a distal portion of driver **60** and introducing driver head **62** into the opening of housing **222**, and (ii) with a second hand, reversibly actuating driver head **62** by operating the adjuster while grasping the distal portion of the driver with the first hand (FIG. 5C; operator's hands not shown). It is hypothesized that this advantageously improves wielding and operation of driver **60** in combination with device **220**.

(60) For some applications, base **300** is coupled to system **120**, e.g., as described hereinabove for device **100** with respect to FIGS. 3A-C, mutatis mutandis.

(61) For some applications, base **300** does not define a handle receptacle, but instead serves only to hold device **220**.

(62) For some applications, device **20**, device **100**, and/or device **220** is used in combination with one or more techniques described in one or more of the following references, which are all incorporated herein by reference: U.S. patent application Ser. No. 12/437,103 to Zipory et al., filed May 7, 2009, which published as US 2010/0286767. For example, (1) device **100** of the present application may be used to facilitate the techniques described with reference to FIGS. 2-3 and/or 6A-12 of US 2010/0286767 to Zipory et al., mutatis mutandis; (2) anchor driver **60** of the present application may comprise or correspond to anchor driver **68** and/or anchor deployment manipulator **24** of US 2010/0286767 to Zipory et al., mutatis mutandis; (3) tissue anchor **40** of the present application may comprise or correspond to anchor **38** of US 2010/0286767 to Zipory et al., mutatis mutandis; and/or (4) implant **140** of the present application may comprise or correspond to annuloplasty ring **22** of US 2010/0286767 to Zipory et al., mutatis mutandis. U.S. patent application Ser. No. 12/689,635 to Zipory et al., filed Jan. 19, 2010, which published as US 2010/0280604. For example, (1) device **100** of the present application may be used to facilitate the techniques described with reference to FIGS. 2-3 and/or 11A-17 of US 2010/0280604 to Zipory et al., mutatis mutandis; (2) anchor driver **60** of the present application may comprise or correspond to anchor driver **68** and/or anchor deployment manipulator **24** of US 2010/0280604 to Zipory et al., mutatis mutandis; (3) tissue anchor **40** of the present application may comprise or correspond to anchor **38** of US 2010/0280604 to Zipory et al., mutatis mutandis; and/or (4) implant **140** of the present application may comprise or correspond to annuloplasty ring **22** of US 2010/0280604 to Zipory et al., mutatis mutandis. PCT patent application IL2012/050451 to Sheps et al., filed Nov. 8, 2013, which published as WO 2013/069019. For example, (1) device **100** of the present application may be used to facilitate the techniques described with reference to FIGS. 14A-I of WO 2013/069019 to Sheps et al., mutatis mutandis; (2) system **120** of the present application may comprise or correspond to system of WO 2013/069019 to Sheps et al., mutatis mutandis; (3) anchor driver **60** of the present application may comprise or correspond to anchor deployment manipulator **61** and/or anchor driver **36** of WO 2013/069019 to Sheps et al., mutatis mutandis; and/or (4) implant **140** of the present application may comprise or correspond to annuloplasty structure **222** and/or sleeve **26** of WO 2013/069019 to Sheps et al., mutatis mutandis. PCT patent application IL2013/050860 to Sheps et al., titled “Controlled steering functionality for implant-delivery tool”, filed on Oct. 23, 2013, which published as WO 2014/064694. For example, (1) device **100** of the present application may be used to facilitate techniques described with reference to FIGS. 10A-I, 12A-14B, 18A-C, 21-28, 34, and 36 of this PCT application titled “Controlled steering functionality for implant-delivery tool”, mutatis mutandis; (2) system **120** of the present application may comprise or correspond to system of this PCT application titled “Controlled steering

functionality for implant-delivery tool”, mutatis mutandis; anchor driver **60** of the present application may comprise or correspond to anchor deployment manipulator **61**, anchor driver **36** and/or anchor driver **2338** of this PCT application titled “Controlled steering functionality for implant-delivery tool”, mutatis mutandis; and/or (4) implant **140** of the present application may comprise or correspond to annuloplasty structure **222** and/or sleeve **26** of this PCT application titled “Controlled steering functionality for implant-delivery tool”, mutatis mutandis. PCT patent application IL2013/050861 to Herman et al., titled “Percutaneous tissue anchor techniques”, filed on Oct. 23, 2013, which published as WO 2014/064695. For example, (1) device **100** of the present application may be used to facilitate the techniques described with reference to FIGS. **9A-C** and/or **13A-D** of this PCT application titled “Percutaneous tissue anchor techniques”, mutatis mutandis; (2) tissue anchor **40** of the present application may comprise or correspond to tissue anchor **40** of this PCT application titled “Percutaneous tissue anchor techniques”, mutatis mutandis; and/or (3) anchor driver **60** of the present application may comprise or correspond to anchor driver **500**, anchor driver **236**, deployment manipulator **261**, or tool **80** of this PCT application titled “Percutaneous tissue anchor techniques”, mutatis mutandis.

(63) Reference is again made to FIGS. **1A-5C**. Typically, the tissue anchor is dimensioned to fit snugly in the anchor-storage zone of the housing. Typically, the tissue anchor (e.g., core **41** thereof) is dimensioned to slide snugly through the channel of the housing, and for some applications this snug sliding prevents tissue-engaging member **44** of the anchor from touching the housing (e.g., the wall of the channel) when the anchor moves through the channel. Typically, at least a portion of the pin is dimensioned to slide snugly through the chamber.

(64) As described hereinabove, for some applications the obstructing portion of the retaining member of devices **20** and **220** obstructs tissue anchor **40** by engaging core **41** of the anchor. The movement of the retaining member in response to the proximally-directed force applied to the anchor typically moves the obstructing portion such that it does not subsequently engage tissue-engaging member **44** of the anchor. For example, for device **20**, portion **29** moves at least partly laterally out of the anchor-storage zone and/or channel **24** (such that member **44** can move past portion **29** without being engaged by it), and for device **220**, portion **229** moves longitudinally out of channel **224** (such that member **44** can move past portion **229** without being engaged by it). It is hypothesized that for some applications this advantageously reduces a likelihood of the anchor-handling device (e.g., the obstructing portion of the retaining member) damaging tissue-engaging member **44**. It is hypothesized that the use of a retaining member that has an obstructing portion that returns to its original position as soon as core **41** has moved past the obstructing portion, does not have this advantageous feature.

(65) It is to be noted that the technique used to inhibit movement of the retaining member of device **20** may be used to inhibit movement of the retaining member of device **220**, mutatis mutandis, and vice versa.

(66) Reference is again made to FIGS. **1A-5C**. It is to be noted that, for both device **20** and device **220**, proximal withdrawal of anchor **40** typically results in sliding of the retaining member (e.g., the pin) in an at least partly proximal direction. This sliding is typically along an axis that is disposed at an angle of less than 30 degrees with respect to a central longitudinal axis of the channel through which the anchor is withdrawn. For example, for device **20** the angle is typically 8-30 degrees (e.g., 10-20 degrees, such as 11-14 degrees), and for device **220** the angle is typically less than 10 degrees, such as 0 degrees—i.e., parallel with the channel.

(67) Reference is again made to FIGS. **1A-5C**. As described hereinabove, the anchor-handling devices allow retrieval of the tissue anchor(s) disposed therein in response to a proximally-directed force that is greater than a threshold force. Typically the threshold force is greater than 300 grams force and/or less than 1500 grams force (e.g., 300-1500 grams force, e.g., 500-1200 grams force, e.g., 500-1000 grams force, such as 600-800 grams force). Tissue anchor **40** typically has a mass of less than 1 g and/or greater than 0.01 g (e.g., 0.01-1 g, e.g., 0.05-0.2 g, e.g., 0.07-0.12 g, such as

about 0.1 g). Thus the threshold force (measured in grams force) is typically greater than 300 times (e.g., greater than 1000 times, e.g., greater than 3000 times, e.g., greater than 10,000 times, such as greater than 15,000 times) and/or less than 150,000 times the mass of the tissue anchor (measured in grams). It is to be noted that the threshold force is therefore many times greater than that which would be required simply to prevent the tissue anchor from undesirably exiting the device due to gravity and/or movement of the device (e.g., during transport).

(68) This configuration of the anchor-handling device serves to test coupling of the anchor driver to the tissue anchor before releasing the tissue anchor. Only if the coupling is sufficient to support a proximally-directed force that is greater than the threshold force, will the device release the anchor. This is hypothesized to increase safety and reliability of the use of the anchor and driver, e.g., by reducing a likelihood that the anchor will subsequently become disengaged from the driver at an undesired time (e.g., within the body of a subject). Whereas one might consider testing the anchor-driver coupling subsequently to removal of the anchor from the anchor-handling device, such post-removal testing requires an extra procedural step, and for some applications it increases a likelihood of damaging and/or contamination of the (typically sterile) tissue anchor. Furthermore, whereas the anchor-handling devices described herein facilitate making a second attempt at coupling the driver to the same anchor, post-removal testing typically does not.

(69) Reference is again made to FIGS. 1A-5C. Typically the anchors are provided sterile within the anchor-handling device. As described hereinabove, for some applications, the anchor-handling device is configured such that returning the exposed portion of the retaining member back into the housing requires a distally-directed force that is more than twice as great (in the opposite direction) as the threshold force that was previously required to move the portion out of the housing. For example, the moving of the portion back into the housing may be in effect prevented. As well as facilitating the exposed portion serving as an empty-housing indicator, this characteristic of the anchor-handling device discourages and/or prevents the operator from returning a previously-removed anchor into the device, e.g., thereby ensuring that only sterile anchors are disposed within the device.

(70) For some applications, the anchor-handling devices described herein are configured to be at least in part submerged in saline prior to and/or during use, e.g., to reduce a likelihood of air (e.g., bubbles) being retained by the anchor and/or driver and subsequently introduced into the subject.

(71) It will be appreciated by persons skilled in the art that the present invention is not limited to what has been particularly shown and described hereinabove. Rather, the scope of the present invention includes both combinations and subcombinations of the various features described hereinabove, as well as variations and modifications thereof that are not in the prior art, which would occur to persons skilled in the art upon reading the foregoing description.

Claims

1. A system, comprising: a tissue anchor; and an anchor-handling device: comprising a housing, shaped to define a channel having an anchor-storage zone and a proximal opening, the anchor stored in the anchor-storage zone, configured such that, while the tissue anchor is stored in the anchor-storage zone, the tissue anchor is movable out of the anchor-storage zone toward the proximal opening in response to a proximally-directed force being applied to the tissue anchor, and comprising an element that serves as an indicator of movement of the tissue anchor out of the anchor-storage zone toward the proximal opening by the element sliding proximally within the housing in response to the proximally-directed force: from an anchor-stored position that is indicative of the presence of the anchor within the anchor-storage zone, into an anchor-absent position that is (i) different from the anchor-stored position, and (ii) indicative of an absence of the anchor from the anchor-storage zone by a part of the element protruding proximally out of the housing.

2. The system according to claim 1, wherein the system is configured to resist returning of the part of the element distally into the housing.
3. The system according to claim 1, wherein: the tissue anchor comprises a coupling head and a tissue-engaging member, the system further comprises an anchor driver comprising an anchor-engaging head, a handle, and a shaft therebetween, and the anchor-engaging head is dimensioned to be advanceable into the anchor-storage zone, and configured to lock to the coupling head within the anchor-storage zone, and the anchor driver is configured to apply the proximally-directed force to the anchor while the anchor-engaging head is locked to the coupling head within the anchor-storage zone.
4. The system according to claim 3, wherein the shaft is flexible and is configured to transluminally advance the tissue anchor into a subject while the anchor-engaging head remains locked to the tissue anchor.
5. The system according to claim 1, wherein: the housing is shaped to define a chamber that is in fluid communication with the channel, and the element is configured to slide within the chamber in response to the proximally-directed force applied to the tissue anchor.
6. The system according to claim 5, wherein: the element is configured to slide proximally within the chamber in response to the proximally-directed force applied to the tissue anchor, the anchor-handling device comprises: a detent; and a cavity, and in response to movement of the element sufficiently proximally (i) to allow the tissue anchor to be removed from the anchor-handling device, and (ii) such that the element transitions into the anchor-absent position, the detent moves into the cavity, thereby inhibiting subsequent returning of the part of the element into the housing by inhibiting distal movement of the element with respect to the chamber.
7. The system according to claim 6, wherein: the element defines the detent, and the housing defines the cavity.
8. The system according to claim 6, wherein: the element defines the cavity, and the anchor-handling device further comprises an inhibitor tongue, the inhibitor tongue defining the detent.
9. The system according to claim 6, wherein: the anchor-stored position is a retaining state in which the element is configured to retain the tissue anchor in the anchor-storage zone, and the element is configured to allow the tissue anchor to leave the anchor-storage zone by moving in response to the proximally-directed force only if the proximally-directed force is greater than a pre-determined threshold force.
10. The system according to claim 9, wherein: the cavity is a first cavity of the anchor-handling device, the anchor-handling device further comprises a second cavity, in the anchor-stored position of the element, the detent is disposed within the second cavity, the disposition of the detent within the second cavity retaining the element in the anchor-stored position by inhibiting proximal movement of the element with respect to the chamber, and the anchor-handling device is configured such that the detent is movable proximally out of the second cavity only in response the proximally-directed force exceeding the threshold force.
11. The system according to claim 9, wherein: the element has a distal part, the part of the element is a proximal part of the element, and wherein: in the anchor-stored position, the distal part engages the tissue anchor, and in response to the proximally-directed force applied to the tissue anchor, the element slides proximally within the chamber and into the anchor-absent position, such that the proximal part of the element protrudes proximally out of the housing.
12. The system according to claim 1, wherein the tissue anchor is in contact with the element such that the proximally-directed force applied to the tissue anchor drags the element proximally, along with the tissue anchor.
13. The system according to claim 12, wherein the element defines a recess and wherein: the tissue anchor resides within the recess, and the system is configured such that once the tissue anchor is exposed out of the proximal opening, the tissue anchor is removable from the recess laterally with respect to the element.

14. Apparatus comprising: a tissue anchor; and an anchor-storage device, comprising: a housing, shaped to define: a cavity, a channel having an anchor-storage zone and a proximal opening, the anchor stored in the anchor-storage zone, and a retaining member that: comprises a detent, has a retaining state in which the retaining member is configured to retain the tissue anchor in the anchor-storage zone, and is transitionable away from the retaining state in response to a proximally-directed force applied to the tissue anchor: to allow the tissue anchor to leave the anchor-storage zone in response to the proximally-directed force, and such that the detent moves proximally and into the cavity to engage the cavity in a manner that inhibits subsequent movement of the retaining member distally within the housing.
15. The apparatus according to claim 14, wherein at least part of the retaining member is elastically deformable.
16. The apparatus according to claim 14 wherein: the cavity is a first cavity, the housing defines a second cavity, and the retaining member is arranged such that, responsively to the proximally-directed force being applied to the tissue anchor, the retaining member deforms and moves proximally with the tissue anchor such that the detent moves from the first cavity into the second cavity.
17. The apparatus according to claim 16, wherein the detent is configured to, when in the second cavity, inhibit subsequent returning of the retaining member distally into the housing.
18. A system, comprising: a tissue anchor; and an anchor-handling device: comprising a housing: defining a cavity, shaped to define a channel having an anchor-storage zone and a proximal opening, the anchor stored in the anchor-storage zone, and comprising a retaining member that: defines a detent, engages the housing by the detent protruding into the cavity in a manner that retains the tissue anchor within the anchor-storage zone, and responsively to a proximally-directed force being applied to the tissue anchor, deforms and moves proximally with the tissue anchor to allow the anchor to move out of the anchor-storage zone and the housing.
19. The system according to claim 18, wherein at least part of the retaining member is elastically deformable.
20. The system according to claim 18, wherein: the cavity is a first cavity, the housing defines a second cavity, and the retaining member is arranged such that, responsively to the proximally-directed force being applied to the tissue anchor, the retaining member deforms and moves proximally with the tissue anchor such that the detent moves from the first cavity into the second cavity.
21. The system according to claim 20, wherein the detent is configured to, when in the second cavity, inhibit subsequent returning of the retaining member distally into the housing.
22. The system according to claim 18, wherein: the tissue anchor comprises a coupling head and a tissue-engaging member, the system further comprises an anchor driver comprising an anchor-engaging head, a handle, and a shaft therebetween, and the anchor-engaging head is dimensioned to be advanceable into the anchor-storage zone, and configured to lock to the coupling head within the anchor-storage zone, and the anchor driver is configured to apply the proximally-directed force to the anchor while the anchor-engaging head is locked to the coupling head within the anchor-storage zone.
23. The system according to claim 22, wherein the shaft is flexible and is configured to transluminally advance the tissue anchor into a subject while the anchor-engaging head remains locked to the tissue anchor.
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